

Biological AI and the honest model scorecard

Better than what, for which task, on which unseen data?

By the end, you should be able to...

Explain an embedding and the difference between zero-shot use and fine-tuning.

Compare a model with a simple baseline under a matched evaluation.

Write a claim limited to the task, data and evidence actually tested.

Reconnect with the biology

Models see numerical representations of biological measurements. A useful representation does not itself establish a mechanism.

Think silently -> explain to a neighbour -> check the measurement.

An embedding is a numerical representation

01

A sequence or cell profile can be represented by a vector of numbers.

>

02

Nearby vectors may share patterns useful for a particular task.

>

03

The numbers have no automatic biological interpretation.

Pretraining and task adaptation are different

Pretraining

Task-specific adaptation

Pretraining uses a broad dataset and a specified objective.

Fine-tuning updates a model for another task using additional data.

Zero-shot use does not perform task-specific model training.

A biological example: scGPT

01

scGPT is one published approach to pretrained single-cell representations.

>

02

Its original study investigated several adapted downstream tasks.

>

03

Those results do not establish success for every tissue or unseen setting.

Independent evaluation can ask a different question

Original task evidence

Independent stress test

Kedzierska et al. tested zero-shot behavior of specified models and datasets.

In some tested settings, simpler approaches performed better.

This is not evidence that all biological foundation models always fail.

A fair comparison fixes the rules

1 Use the same task, eligible inputs, evaluation split and metric.

2 Include a simple baseline and explain training-data overlap where known.

3 Separate validation choices from the final test.

Record the score and what it leaves out

Task + data

Split + adaptation

Baseline + metric

Limits + uncertainty

Performance on held-out independent units.

Data coverage, uncertainty, computation and failures.

A stronger number on one dataset does not prove transfer to another.

A virtual cell is a claim to unpack

01

Which intervention, biological scale and outcome are modeled?

>

02

Which predictions were checked experimentally or on independent data?

>

03

What is outside the validated scope?

Use the least complex sufficient method

Complexity adds dependencies and opportunities for hidden failure.

A small baseline is an informative comparator, not an embarrassment.

Select the method supported by the evaluation and constraints.

Worked example: a result with honest scope

A fictional scorecard gives baseline accuracy 0.78 and model accuracy 0.82 on 50 held-out donors.

Predict the result and one limitation before calculating.

Work through the reasoning

The observed difference is 0.04 on this test set.

Uncertainty and paired errors are needed before declaring a reliable improvement.

The result does not establish performance in another tissue or hospital.

Pause: make a defensible claim

Model A uses leaked donor rows and reports 0.95; model B holds out donors and reports 0.75. Can their scores rank the models fairly?

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

No. The evaluation targets differ. Re-evaluate on matched independent test units before comparing performance.

Today's practical

W11 studio: complete a six-field model scorecard from supplied fictional results and two verified publication evidence cards. No model training or GPU is needed.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

1 embedding

2 pretraining

3 fine-tuning

4 zero-shot

5 baseline

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: sketch a domain-shift stress test without implementing a foundation model.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

S11 scikit-learn: Common pitfalls and recommended practices.

S14 Cui et al. (2024). scGPT: toward building a foundation model for single-cell multi-omics using generative AI.

S15 Kedzierska et al. (2025). Zero-shot evaluation reveals limitations of single-cell foundation models.